REALIZATION CALCULUS

# Komorebi Physics

## Continuum, Geometry and Refinement

FIELDS / STRUCTURE / LIMITS

Komorebi P.

2026

KOMOREBI PHYSICS

# Komorebi Physics

## Continuum, Geometry and Refinement

Komorebi P.

A mathematical account of resolved dynamics, free fields, gauge transport, gravity, arithmetic structure, and cosmological models.

2026

CONTENTS

# Mathematical constructions


## I Resolution and free dynamics

Projectors, Packets, and Transfer 2
Resolution, conservation, and operator reconstruction

Fourier Packets and the Continuum Dirac Operator 35
Clifford matrices, Fourier completion, and the exact domain

The Dirac Group and Its Generator 67
Spectral evolution, resolvents, and regularity-dependent rates

## II Gauge transport and gravity

Gauge Covariance and Geometric Refinement 75
Ordered transport, conservation, and curvature estimates

Constrained Gravity and Strong Fields 107
Canonical structure, local reduction, and spherical tests

## III Arithmetic and polyphase refinement

Adic and Polyphase Quantum Refinement 161
Haar packets, spectral reduction, and infinite hierarchies

## IV Matter and cosmological transport

Covariant Refinement-Fiber Dynamics 213
Symmetry, stress exchange, production, and ultraviolet bounds

Cosmology, Dark Sectors, and Structure Formation 245
Conditional transport, structure, and linked rejection tests

PART I

# Resolution and free dynamics

Exact packet algebra becomes field dynamics only after reconstruction, completion, and domain control are supplied. The argument moves from finite transfers to a torus Hamiltonian and then identifies its full unitary group.

IN THIS PART


Projectors, Packets, and Transfer 2

Fourier Packets and the Continuum Dirac Operator 35

The Dirac Group and Its Generator 67

# Projectors, Packets, and Transfer

Resolution, conservation, and the passage to infinite systems

## Resolution, dynamics, and reconstruction

A physical description often begins by separating a state into spatial cells, frequency bands, spin channels, tensor factors, or spectral sectors. Such a separation is useful only when the components can be reconstructed, their interactions can be described, and different resolutions can be compared. Those requirements lead to the calculus developed here.

An orthogonal projector selects a packet. Placing an operator between two projectors gives a transfer block. A skew-adjoint generator determines an antisymmetric current between packet weights. Grouping the labels produces a coarser description in which internal currents cancel. These statements can be understood in a small matrix example before any infinite-dimensional analysis is needed.

The same questions become more demanding when frames move, states become mixed, or the system exchanges information with an environment. Countable resolutions introduce convergence. Unbounded operators introduce domains. Each extension retains part of the finite algebra and adds a hypothesis that must remain visible where it is used.

The free Dirac construction that follows gives these ideas a concrete analytic realization. Fourier modes resolve a spinor field, finite matrices govern its components, and completion leads to a self-adjoint Hamiltonian. Gauge transport, constrained gravity, and arithmetic refinement then test which parts of the same language survive in different settings.

> **The organizing question**
>
> Which quantities can be reconstructed exactly, which balances survive a change of resolution, and which additional hypotheses are required for an infinite or unbounded system?

The mathematical arguments use ordinary complex linear algebra, operator theory, and Hilbert-space analysis. Formal source correspondences and verification records are collected in the technical appendices. Numerical illustrations explain specified examples rather than replacing universal proofs.

Then $\psi_A = (1, i, 0, 0)^{\mathsf{T}}/\sqrt{7}$ and $\psi_B = (0, 0, 2, -i)^{\mathsf{T}}/\sqrt{7}$. The weights are $p_A = 2/7$ and $p_B = 5/7$. Refining into the four coordinate projectors gives weights $1/7, 1/7, 4/7, 1/7$. The two descriptions resolve the same vector at different levels. The coarse description still contains the two-dimensional packet vectors. Retaining only the two scalar weights would discard their internal phases and amplitudes.

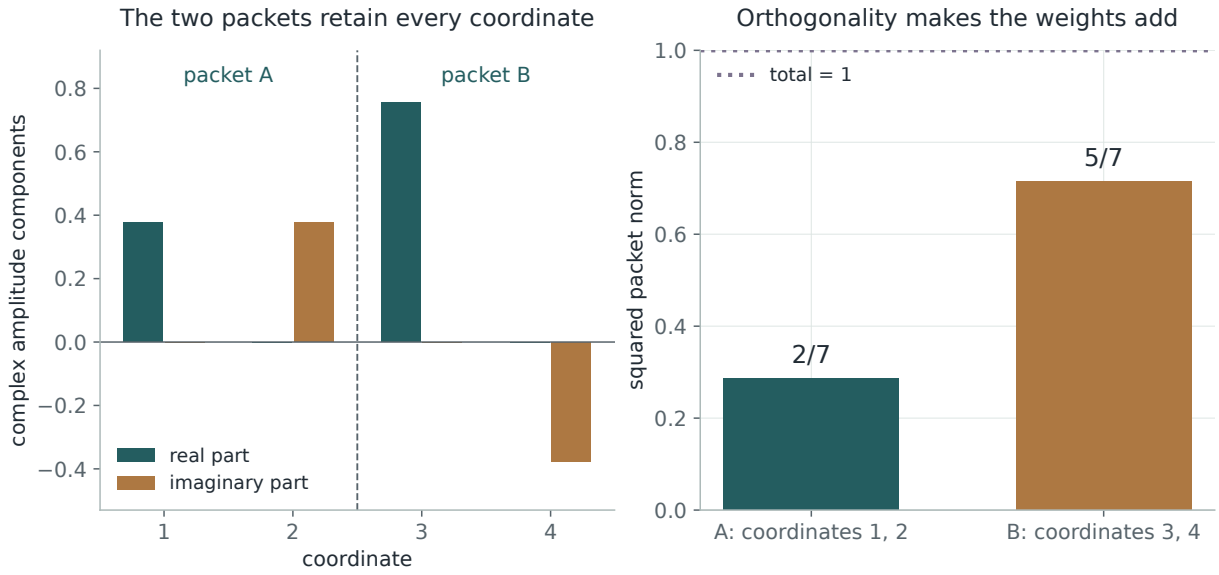



**Figure 1.** The four-coordinate state of Example 1.3. Packet vectors retain amplitudes and phases, while their two weights retain only the squared sizes.

## 2 Operator blocks and exact reconstruction

The same atlas resolves an operator on both its input and output sides. For $X \in \mathcal{B}(\mathcal{H})$, define

$$X_{ba} = P_b X P_a. \tag{3}$$

The first index names the destination and the second names the source. As a map between packet spaces, this is $X_{ba} : \operatorname{Ran} P_a \to \operatorname{Ran} P_b$. As a bounded operator on $\mathcal{H}$, it is extended by zero on the orthogonal complement of $\operatorname{Ran} P_a$. Keeping these two descriptions aligned prevents a block from being composed with an incompatible input space.

**Proposition 2.1** (Block calculus)**.** *For bounded $X, Y$,*

$$X = \sum_{b,a} X_{ba}, \qquad P_b X \psi = \sum_a X_{ba} \psi_a, \tag{4}$$

$$(XY)_{ca} = \sum_b X_{cb} Y_{ba}, \qquad (X^*)_{ab} = (X_{ba})^*. \tag{5}$$

*The identity has diagonal blocks $P_a$ and zero off-diagonal blocks. If $X = X^*$, its opposite blocks are adjoints.*

# Fourier Packets and the Continuum Dirac Operator

From finite spinor dynamics to a self adjoint Hamiltonian on the three torus

## From finite spinors to a continuum Hamiltonian

The preceding calculus becomes a concrete field construction when Fourier modes are used as packets. Each mode carries a four-component spinor and an explicit Hermitian matrix. The field is reconstructed through an isometric Fourier map. Exact norm preservation, completion, approximation, and control of the adjoint domain justify the passage from finite calculations to continuum analysis.

The setting is a free spinor on a fixed flat three-torus. For every real mass parameter, the resulting Hamiltonian is densely defined, self-adjoint, and closed. Its domain is exactly the first-order weighted Fourier square-summability domain. These assertions include the operator domain, not merely a formal differential expression.

The Hamiltonian and the modewise unitary evolution are first constructed separately. The following chapter, *The Dirac Group and Its Generator*, connects them by proving strong continuity, equality of the full generator domain with the Hamiltonian domain, and the spectral-exponential identity. It also develops graph-domain and strong-resolvent convergence for the centered Fourier family and a quadratic spacing estimate under third-order Fourier regularity.

> **The role of this construction**
>
> Finite Clifford identities determine each mode. Hilbert-space completion reconstructs the field. Adjoint-domain equality determines the self-adjoint operator. The generator argument then identifies its evolution. These are distinct steps in one continuous mathematical construction.

The operator construction, full generator-domain identification, centered graph and strong-resolvent limits, and third-order moment estimates have recorded Lean verification. The abstract spectral-functional-calculus identity remains a written deduction without a separate formal API linkage. The technical appendices identify the exact formal statements and evidence. None of these mathematical results by itself identifies an observed particle or supplies a dynamical spacetime geometry.

Define the unitary family previously constructed mode by mode:

$$(U(t)f)_n = \exp(-itH_m(n))f_n. \tag{5}$$

As in the earlier modules, the complex inner product is linear in its second coordinate. Equations (3) and (5) fix the sign of the generator to be $-iH$.

### 1.1 Why the distinction matters

An exact matrix identity on each mode need not automatically justify an infinite dimensional derivative, because the norms $\|H_m(n)\|$ grow with $|n|$. Nor does equality on finite Fourier packets alone identify the maximal generator. The next proof controls the tails through the natural graph norm. This supplies the missing step between the checked propagator and the unbounded Hamiltonian.

## 2 The continuum group and its maximal generator

**Lemma 2.1** (Strong continuity)**.** *The family $(U(t))_{t\in\mathbb{R}}$ is a strongly continuous unitary group on $\mathcal{H}$. Its action commutes with every finite Fourier truncation.*

*Proof.* Every $H_m(n)$ is Hermitian, so its exponential is unitary. The group law and $U(0) = \mathrm{I}$ hold at each $n$ and hence in $\mathcal{H}$. For any finite $S \subset \mathbb{Z}^3$, let $P_S$ be coefficient truncation. Diagonality gives $U(t)P_S = P_S U(t)$ exactly. For $f \in \mathcal{H}$, select $S$ so that $\|f - P_S f\| < \varepsilon/4$. At any $t$,

$$\|U(t)f - f\| \le 2\,\|f - P_S f\| + \|U(t)P_S f - P_S f\|\,.$$

The second term tends to zero as $t \to 0$, being the norm of a finite tuple of continuous matrix exponentials. The first is smaller than $\varepsilon/2$. Thus $U(t)f \to f$ for every $f$. Continuity at other times follows from the group law and unitarity. □

**Lemma 2.2** (Domain preservation and graph isometry)**.** *For every $t \in \mathbb{R}$, $U(t)\mathcal{D}(H) = \mathcal{D}(H)$, and on this domain*

$$HU(t)f = U(t)Hf, \qquad \|U(t)f\|^2 + \|HU(t)f\|^2 = \|f\|^2 + \|Hf\|^2\,.$$

*Proof.* The exponential of a fixed matrix commutes with that matrix. Consequently $H_m(n)U_n(t)f_n = U_n(t)H_m(n)f_n$. Since $U_n(t)$ preserves the spinor norm, $\sum_n \|H_m(n)U_n(t)f_n\|^2 = \sum_n \|H_m(n)f_n\|^2$. The series is finite precisely on $\mathcal{D}(H)$. Replacing $t$ by $-t$ gives equality of domains, while equality of the two graph norms follows from modewise unitarity. □

**Theorem 2.3** (Maximal strong generator)**.** *The strong generator $\mathsf{A}$ of $U(t)$ has domain exactly $\mathcal{D}(H)$ and acts as $\mathsf{A} = -iH$:*

$$\mathcal{D}(\mathsf{A}) = \mathcal{D}(H), \quad \lim_{t\to 0}\left\|\frac{U(t)f - f}{t} + iHf\right\| = 0 \quad (f \in \mathcal{D}(H)). \tag{6}$$

*Conversely, if the strong difference quotient has a limit for $f \in \mathcal{H}$, then $f \in \mathcal{D}(H)$ and that limit equals $-iHf$.*

*Proof.* Fix $f \in \mathcal{D}(H)$. For each $n$, the ordinary finite dimensional exponential identity gives

$$\frac{U_n(t) - \mathrm{I}}{t} f_n = -i\int_0^1 U_n(st)H_m(n)f_n\,ds \quad (t \neq 0). \tag{7}$$

# Gauge Covariance and Geometric Refinement

Ordered transport, curvature, and the control of changing resolution

## Transport beyond the free field

The free Dirac construction resolves a translation-invariant field into independent Fourier modes. Gauge coupling changes that organization. Transport between neighboring fibers carries additional information, and a comparison between resolutions must preserve the order and orientation of the links.

A link has a named source and target. Its transformation law is fixed by that direction. Composition cancels the gauge factors at internal endpoints, while a closed path retains conjugation at its basepoint. This elementary mechanism supports the treatment of holonomy, curvature, covariant differences, and refinement.

The finite Maxwell–Dirac example combines matter continuity, electric work, and propagation of the Gauss constraint. Geometric interpolation then separates edge compatibility, curvature defects, and Hilbert normalization. Regularity assumptions turn these structural conditions into quantitative estimates.

The constructions concern specified finite systems and prescribed background connections. A coupled continuum gauge theory additionally requires an operator domain, existence of the field evolution, propagation of constraints, and a convergence theorem in an explicitly chosen topology. Local covariance does not discharge those obligations.

The visual atlas develops the finite examples. The technical appendix records formal source coverage and the limits of its verification evidence. The argument here remains readable without following implementation identifiers.

The spectral index is the logarithmic derivative of the power law:

$$\frac{\mathrm{d}\log \mathcal{P}_{\mathcal{R}}}{\mathrm{d}\log k} = 3 - 2\nu. \tag{5.35}$$

Using $\nu = (3-\epsilon)/[2(1-\epsilon)]$ gives $3 - 2\nu = -2\epsilon/(1-\epsilon)$. The scalar and tensor spectra have the same $k$-power, so their runnings vanish. Dividing their normalization constants gives

$$r = \frac{2^{2\nu}}{2^{2\nu-4}/\epsilon} = 16\epsilon, \tag{5.36}$$

and substitution of $n_t$ gives the exact consistency relation. The first-order slow-roll relation $r = -8n_t + O(\epsilon^2)$ follows.

## 5.4 Controlled benchmark

For

$$\epsilon = 0.01, \qquad N = 60, \tag{5.37}$$

The power-law calculation gives

$$q = 1.010101, \tag{5.38}$$
$$\nu = 1.510101, \tag{5.39}$$
$$n_s = 0.979798, \tag{5.40}$$
$$n_t = -0.020202, \tag{5.41}$$
$$r = 0.16, \tag{5.42}$$
$$\frac{|\Omega_K|}{|\Omega_{K,i}|} = 2.545749 \times 10^{-52}, \tag{5.43}$$
$$\frac{(aH)^{-1}}{(a_i H_i)^{-1}} = 1.595540 \times 10^{-26}. \tag{5.44}$$

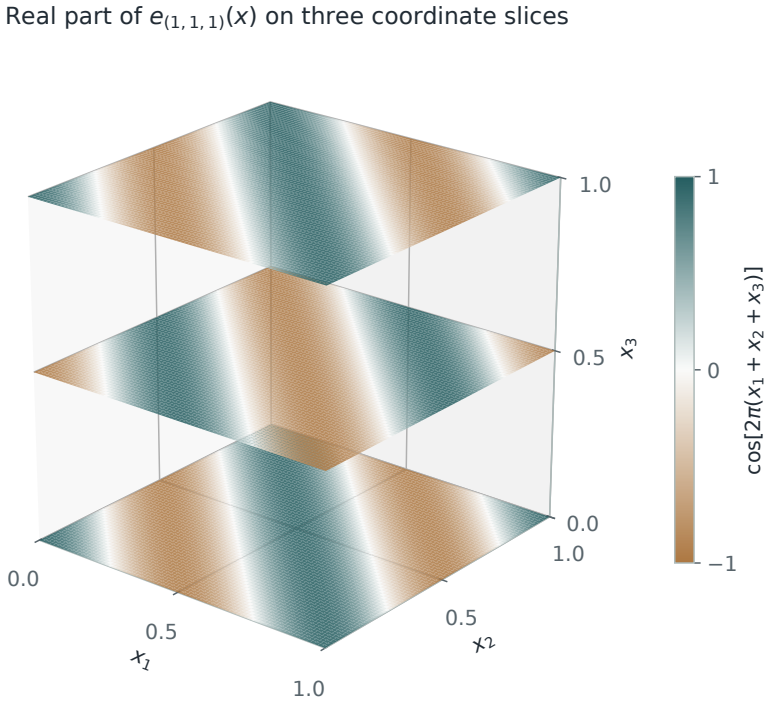
The numerical checks are

$$\varepsilon_{\epsilon_H} = 0, \tag{5.45}$$
$$\varepsilon_{z''/z} = 0, \tag{5.46}$$
$$\varepsilon_{n_s} = 1.179612 \times 10^{-16}, \tag{5.47}$$
$$\varepsilon_{n_t} = 2.255141 \times 10^{-16}, \tag{5.48}$$
$$\varepsilon_r = 2.775558 \times 10^{-17}, \tag{5.49}$$
$$\varepsilon_{\mathrm{W}} = 3.907985 \times 10^{-14}, \tag{5.50}$$
$$p_{v,\mathrm{mid}} = 1.964609. \tag{5.51}$$

## 5.5 Open extensions

- Derive the conditions that select the inflationary attractor from general initial data.
- Construct a graceful-exit mechanism and a controlled reheating sector.
- Determine the vacuum from projector refinement instead of fixing it by hand.
- Quantize the projector geometry together with the scalar perturbations.
- Extend the calculation to multifield dynamics and the leading non-Gaussian observables.

## 3 A Fourier mode lives on the three-torus




Three genuine three-dimensional slices of the real part of $e_{(1,1,1)}(x)$. Equal values on opposite faces reflect periodicity. Color shows a real component, not probability density.

The scalar mode $e_n(x) = \exp(2\pi i n \cdot x)$ has absolute value one everywhere. For normalized Haar measure, its squared $L^2$ norm is also one. Distinct integer modes are orthogonal:

$$\int_{\mathbb{T}^3} \overline{e_n(x)} e_k(x)\, dx = \delta_{nk}.$$

The plate shows the mode $(1, 1, 1)$ at $x_3 = 0, 1/2, 1$. Its real part changes sign, while the magnitude of the full complex wave stays fixed. The upper and lower faces have the same pattern because adding one to a coordinate changes the phase by an integer multiple of $2\pi$.

A spinor amplitude $a_n \in \mathbb{C}^4$ multiplies this scalar mode. For a finite sum $f(x) = \sum_n e_n(x) a_n$, orthogonality removes the cross terms only after integration. It gives $\|f\|_{L^2}^2 = \sum_n \|a_n\|^2$. Local interference can still produce a nonconstant density. This distinction is essential when reading the evolution plate.

# Mathematics, evidence, and interpretation

## The scope of a proof

A theorem asserts what follows from its stated hypotheses. A conditional theorem remains a theorem under those hypotheses. Numerical evidence concerns the declared inputs, tolerances, and implementation. A physical model adds an identification of mathematical objects with quantities in nature. These roles are kept separate throughout the volume.

## The free Dirac construction

The formal record covers mode algebra, completed coefficient and torus evolution, the maximal Hamiltonian, its exact first-order Fourier domain, physical self-adjointness, and the full strong generator. The centered extension proves graph-domain convergence, actual nonreal resolvent inverses and strong limits, and quadratic estimates under a finite third-order Fourier moment. The abstract spectral-functional-calculus identity remains a written deduction without a separate formal API linkage.

## The comparison spaces matter

The centered operators act on the same infinite torus Fourier index space. A changing finite spatial grid additionally needs sampling, interpolation, and aliasing control. A background coupling changes the operator-domain problem. A dynamical gauge or gravitational field requires further evolution and constraint arguments.

## Reading the technical appendices

The detailed companions identify exact declarations, hypotheses, source records, computations, and missing corroboration. A recorded compiler pass is not a new independent compilation. A checksum validates identity relative to a reference file, not the physical assumptions or the truth of every surrounding interpretation.